1 Microscopic Foundation: Neural Dynamics

We consider a minimal circuit consisting of a single presynaptic neuron connected to a single postsynaptic neuron through a unidirectional synapse of weight w. This section specifies the dynamics governing the postsynaptic membrane potential, the synaptic coupling, and the generation of spikes.

1.1 Membrane Potential Dynamics

The postsynaptic neuron is modeled as a Leaky Integrate-and-Fire (LIF) unit, a standard reduction of the Hodgkin–Huxley formalism [**hodgkinQuantitativeDescriptionMembrane1952**] that retains sub-threshold integration and thresholding [1]. Its membrane potential $V(t)$ evolves according to the conservation of current across the cell membrane, modeled as a parallel RC circuit with leakage resistance R_m and membrane capacitance C_m [2]:

$$\tau_m \frac{dV(t)}{dt} = -(V(t) - E_L) + R_m I_{\text{syn}}(t) + R_m I_{\text{ext}}(t), \tag{1}$$

where $\tau_m = R_m C_m$ is the membrane time constant, E_L is the resting potential, $I_{\text{syn}}(t)$ is the synaptic current from the presynaptic neuron, and $I_{\text{ext}}(t)$ is an external current, typically modeled as an injected current or Gaussian white noise.

1.2 Spike Generation

A spike is defined as the instant t^k_{post} at which the membrane potential crosses a fixed threshold θ from below:

$$t^k_{\text{post}} : \quad V(t^k_{\text{post}}) = \theta \quad \text{and} \quad \left.\frac{dV}{dt}\right|_{t=t^k_{\text{post}}} > 0. \tag{2}$$

The derivative condition distinguishes the upward threshold crossing from subsequent repolarization. Upon firing, V is reset to $V_{\text{reset}} < \theta$ and the dynamics in (1) are suspended for a refractory period τ_{ref}, after which integration resumes from V_{reset} [2].

The spike trains of both neurons are written as sums of Dirac distributions:

$$\rho_{\text{pre}}(t) = \sum_k \delta(t - t^k_{\text{pre}}), \qquad \rho_{\text{post}}(t) = \sum_k \delta(t - t^k_{\text{post}}). \tag{3}$$

The presynaptic spike times $\{t^k_{\text{pre}}\}$ are taken as given (e.g., drawn from a Poisson process), while the postsynaptic times $\{t^k_{\text{post}}\}$ are determined by (1) and (2).

1.3 Synaptic Interaction

The synaptic current is determined by the presynaptic spike train filtered through a postsynaptic current (PSC) kernel and scaled by the synaptic weight w. Under the *current-based* approximation, which treats synaptic currents as independent of the postsynaptic membrane potential, the synaptic current is [1]:

$$I_{\text{syn}}(t) = w \int_{-\infty}^{t} \alpha(t - s)\, \rho_{\text{pre}}(s)\, ds, \tag{4}$$

where $\alpha(t) = \tau_s^{-1} e^{-t/\tau_s}\,\Theta(t)$ is an exponential PSC kernel with synaptic time constant τ_s and $\Theta(t)$ the Heaviside step function [2, 3].

2 Three-Factor Plasticity Model

We now turn to the evolution of the synaptic weight w. The plasticity rule belongs to the class of *three-factor learning rules* reviewed by Frémaux and Gerstner [4] and Gerstner et al. [5]. In standard Spike-Timing-Dependent Plasticity (STDP), weight changes depend on the correlation of pre- and postsynaptic spike times; three-factor rules gate this local signal with a global neuromodulatory factor representing reward or error feedback.

2.1 Weight-Dependent Scaling and Stability

Before specifying the plasticity dynamics, we define the weight-dependent scaling functions that appear in the learning rule. The weight w represents the efficacy of the synapse from the presynaptic to the postsynaptic neuron and is constrained to the interval $[0, w_{\max}]$, where $w_{\max}$ is a physiological saturation limit.

The amplitudes of potentiation and depression are modulated by soft-bound functions [1]:

$$A_+(w) = \eta_+(w_{\max} - w), \qquad A_-(w) = \eta_- w, \tag{5}$$

where η_+ and η_- are learning rates. These linear dependencies cause the rate of weight change to diminish as w approaches either boundary. This prevents unbounded growth of the weight and biases the dynamics toward the interior of $[0, w_{\max}]$. However, because the full weight update (defined below in (10)) involves a signed modulation signal, the soft bounds alone do not strictly guarantee $w \in [0, w_{\max}]$; in practice, a hard clipping step $w \leftarrow \max(0, \min(w, w_{\max}))$ may be applied after each update.

2.2 Local Dynamics: The Eligibility Trace

A central feature of this model is that coincident pre- and postsynaptic spikes create a temporary memory called the *eligibility trace* $E(t)$ [4, 5]. This trace allows the synapse to bridge the temporal gap between millisecond-scale neural activity and delayed reward signals. It evolves as:

$$\tau_e \frac{dE(t)}{dt} = -E(t) + S(t), \tag{6}$$

where τ_e is a decay time constant, typically on the order of hundreds of milliseconds to seconds for reinforcement learning tasks [5].

The driving term $S(t)$ captures the instantaneous STDP induction. To define it, we introduce filtered spike-history variables for each neuron:

$$\tau_+ \frac{dx_{\text{pre}}(t)}{dt} = -x_{\text{pre}}(t) + \rho_{\text{pre}}(t), \tag{7}$$

$$\tau_- \frac{dy_{\text{post}}(t)}{dt} = -y_{\text{post}}(t) + \rho_{\text{post}}(t), \tag{8}$$

where τ_+ and τ_- set the widths of the potentiation and depression windows, respectively. Experimental measurements place these values in the range 20–40 ms [6]. The STDP induction

term then combines Long-Term Potentiation (LTP) and Long-Term Depression (LTD):

$$S(t) = \underbrace{A_+(w)\, x_{\text{pre}}(t)\, \rho_{\text{post}}(t)}_{\text{LTP}} - \underbrace{A_-(w)\, y_{\text{post}}(t)\, \rho_{\text{pre}}(t)}_{\text{LTD}}, \tag{9}$$

with $A_+(w)$ and $A_-(w)$ as defined in (5).

2.3 Global Dynamics: Neuromodulated Update

The weight evolves under the product of the eligibility trace and a global neuromodulatory signal $M(t)$:

$$\frac{dw(t)}{dt} = M(t)\, E(t). \tag{10}$$

To construct $M(t)$, we first define smooth estimates of the instantaneous firing rates by low-pass filtering the spike trains with a rate time constant τ_r:

$$\tau_r \frac{dr_{\text{pre}}(t)}{dt} = -r_{\text{pre}}(t) + \rho_{\text{pre}}(t), \tag{11}$$

$$\tau_r \frac{dr_{\text{post}}(t)}{dt} = -r_{\text{post}}(t) + \rho_{\text{post}}(t). \tag{12}$$

Unlike a rectangular sliding-window estimator, these exponentially weighted averages are smooth functions of time and consistent in form with the other filtered quantities in the model.

As a simplified stand-in for a more general objective, we define the instantaneous reward as a penalty on the deviation of the postsynaptic rate from half the presynaptic rate:

$$R(t) = -\left(r_{\text{post}}(t) - \tfrac{1}{2}\, r_{\text{pre}}(t)\right)^2. \tag{13}$$

The modulation signal $M(t)$ is then a Reward Prediction Error (RPE), computed as the difference between $R(t)$ and a slowly adapting baseline $\bar{R}(t)$ [4, 5]:

$$M(t) = R(t) - \bar{R}(t), \tag{14}$$

where $\bar{R}(t)$ tracks the running average of the reward:

$$\tau_{\bar{R}} \frac{d\bar{R}(t)}{dt} = -\bar{R}(t) + R(t). \tag{15}$$

The baseline enables bidirectional regulation: performance better than expected yields $M > 0$ (reinforcing the current eligibility trace), while performance worse than expected yields $M < 0$ (weakening it).

2.4 Reduced System of Equations

The state variables are $V(t)$, $w(t)$, $x_{\text{pre}}(t)$, $y_{\text{post}}(t)$, $E(t)$, $r_{\text{pre}}(t)$, $r_{\text{post}}(t)$, and $\bar{R}(t)$. The presynaptic spike train $\rho_{\text{pre}}(t) = \sum_k \delta(t - t^k_{\text{pre}})$ is a given input; the postsynaptic spike train $\rho_{\text{post}}(t) = \sum_k \delta(t - t^k_{\text{post}})$ is determined by the threshold crossing $V(t^k_{\text{post}}) = \theta$ with $dV/dt > 0$, followed by reset $V \to V_{\text{reset}}$ and a refractory hold of duration τ_{ref}.

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m\, w \int_{-\infty}^{t} \frac{1}{\tau_s}\, e^{-(t-s)/\tau_s}\, \Theta(t-s)\, \rho_{\text{pre}}(s)\, ds, \tag{16}$$

$$\tau_+ \frac{dx_{\text{pre}}}{dt} = -x_{\text{pre}} + \rho_{\text{pre}}(t), \tag{17}$$

$$\tau_- \frac{dy_{\text{post}}}{dt} = -y_{\text{post}} + \rho_{\text{post}}(t), \tag{18}$$

$$\tau_e \frac{dE}{dt} = -E + \eta_+ (w_{\max} - w)\, x_{\text{pre}}\, \rho_{\text{post}}(t) - \eta_-\, w\, y_{\text{post}}\, \rho_{\text{pre}}(t), \tag{19}$$

$$\tau_r \frac{dr_{\text{pre}}}{dt} = -r_{\text{pre}} + \rho_{\text{pre}}(t), \tag{20}$$

$$\tau_r \frac{dr_{\text{post}}}{dt} = -r_{\text{post}} + \rho_{\text{post}}(t), \tag{21}$$

$$\tau_{\bar{R}} \frac{d\bar{R}}{dt} = -\bar{R} - \left(r_{\text{post}} - \tfrac{1}{2}\, r_{\text{pre}}\right)^2, \tag{22}$$

$$\frac{dw}{dt} = \left[-\left(r_{\text{post}} - \tfrac{1}{2}\, r_{\text{pre}}\right)^2 - \bar{R}\right] E, \tag{23}$$

2.5 Inter-Spike Dynamics

Between spikes, all Dirac delta terms vanish ($\rho_{\text{pre}} = \rho_{\text{post}} = 0$) and the system reduces to a set of autonomous, decoupled relaxation equations. Let $I_s(t)$ denote the residual synaptic current inherited from past presynaptic spikes; between spikes it obeys $dI_s/dt = -I_s/\tau_s$. The inter-spike dynamics are:

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m\, w\, I_s(t), \tag{24}$$

$$\frac{dx_{\text{pre}}}{dt} = -\frac{x_{\text{pre}}}{\tau_+}, \tag{25}$$

$$\frac{dy_{\text{post}}}{dt} = -\frac{y_{\text{post}}}{\tau_-}, \tag{26}$$

$$\frac{dE}{dt} = -\frac{E}{\tau_e}, \tag{27}$$

$$\frac{dr_{\text{pre}}}{dt} = -\frac{r_{\text{pre}}}{\tau_r}, \tag{28}$$

$$\frac{dr_{\text{post}}}{dt} = -\frac{r_{\text{post}}}{\tau_r}, \tag{29}$$

$$\tau_{\bar{R}} \frac{d\bar{R}}{dt} = -\bar{R} - \left(r_{\text{post}} - \tfrac{1}{2}\, r_{\text{pre}}\right)^2, \tag{30}$$

$$\frac{dw}{dt} = \left[-\left(r_{\text{post}} - \tfrac{1}{2}\, r_{\text{pre}}\right)^2 - \bar{R}\right] E. \tag{31}$$

The trace and rate variables (25)–(29) decay exponentially toward zero with their respective time constants, receiving no new input in the absence of spikes. The eligibility trace (27) likewise

decays, since the STDP induction term $S(t)$ vanishes identically when both spike trains are silent. The reward baseline (30) relaxes toward the current instantaneous reward, which itself evolves smoothly as the rate estimates decay.

2.6 Summary of Symbols

Table 1: Summary of notation.

Symbol	Description	Equation
$V(t)$	Postsynaptic membrane potential	(1)
E_L	Resting (leak) potential	(1)
θ	Spike threshold	(2)
V_{reset}	Reset potential after spike	§1.2
$\rho_{\text{pre}}(t),\ \rho_{\text{post}}(t)$	Spike trains (sums of Dirac deltas)	(3)
$\alpha(t)$	Postsynaptic current kernel	(4)
$I_{\text{syn}}(t)$	Total synaptic current	(4)
$I_{\text{ext}}(t)$	External / noise current	(1)
w	Synaptic weight	(10)
$A_+(w),\ A_-(w)$	Weight-dependent LTP/LTD amplitudes	(5)
$x_{\text{pre}}(t)$	Presynaptic spike-history trace	(7)
$y_{\text{post}}(t)$	Postsynaptic spike-history trace	(8)
$S(t)$	STDP induction term	(9)
$E(t)$	Eligibility trace	(6)
$r_{\text{pre}}(t),\ r_{\text{post}}(t)$	Exponentially filtered firing rates	(11)–(12)
$R(t)$	Instantaneous reward signal	(13)
$\bar{R}(t)$	Reward baseline (running average)	(15)
$M(t)$	Neuromodulatory signal (RPE)	(14)

2.7 Parameter Values

Table 2: Model parameters and representative values. Ranges are drawn from the cited experimental and modeling literature.

Parameter	Description	Typical value	Source
τ_m	Membrane time constant	10–20 ms	[2]
R_m	Membrane resistance	10–100 MΩ	[2]
C_m	Membrane capacitance	τ_m/R_m	—
E_L	Resting potential	−70 mV	[2]
θ	Spike threshold	−55 mV	[2]
V_{reset}	Reset potential	−70 mV	[2]
τ_{ref}	Absolute refractory period	2–5 ms	[1]
τ_s	Synaptic time constant (PSC)	2–10 ms	[3]
$w_{\max}$	Maximum synaptic weight	model-dependent	—
η_+	LTP learning rate	model-dependent	—
η_-	LTD learning rate	model-dependent	—
τ_+	Potentiation window width	20–40 ms	[6]
τ_-	Depression window width	20–40 ms	[6]
τ_e	Eligibility trace decay	0.1–1.0 s	[5]
τ_r	Rate-estimation time constant	50–200 ms	—
$\tau_{\bar{R}}$	Reward baseline time constant	1–10 s	[4]

References

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